

STT 861: Theory of Probability & Statistics I

Lecture 3: Combinations & Probability Axioms

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Reference: Casella & Berger Ch. 1.2

Lecture 3 Outline

Today

- Unordered sampling – Combinations
- Probability: formal definition (Kolmogorov axioms)
- σ -algebras (Borel fields)

Unordered Sampling – Combinations

Combination – $\binom{N}{n}$

The number of ways to choose n objects from N *without replacement* and *without regard to order* is:

$$\binom{N}{n} = \frac{N!}{n!(N-n)!} = \frac{P(N, n)}{n!}$$

Read: “ N choose n ”. Also written $C(N, n)$ or C_n^N .

Why divide by $n!$?

Each unordered set of n objects corresponds to $n!$ ordered arrangements.

Committee Example

Choose committee of 3 from 10 people:

$$\binom{10}{3} = \frac{10!}{3!7!} = \frac{720}{6} = \mathbf{120}$$

Key Identity: Pascal's Rule

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

Proof: Consider element a_1 .

Groups containing a_1 : $\binom{n-1}{k-1}$

Groups not containing a_1 : $\binom{n-1}{k}$

Unordered Sampling Without Replacement

Motivating Example: $N = 4$, $n = 2$

Choose 2 from $\{A, B, C, D\}$ *without* replacement. Ordered pairs:

$$\underbrace{(A, B), (A, C), (A, D), (B, C), (B, D), (C, D)}_6 + \underbrace{(B, A), (C, A), (D, A), (C, B), (D, B), (D, C)}_6$$

12 = total ordered pairs; 6 = unordered selections (combinations).

Relation: ordered vs unordered

$$\underbrace{(N)_n}_{\text{ordered}} = \underbrace{c}_{\text{combos}} \times \underbrace{n!}_{\text{orderings of each group}}$$
$$\Rightarrow c = \frac{(N)_n}{n!} = \frac{N!}{n! (N - n)!} = \binom{N}{n} = \binom{N}{N - n}$$

Read $\binom{N}{n}$ as “ **N choose n** ”.

The Binomial Coefficient

Definition: N Choose n

$$\binom{N}{n} = \frac{N!}{n! (N - n)!}$$

Conventions:

- $0! = 1$
- $\binom{N}{0} = 1$
- $\binom{N}{N} = 1$
- $\binom{N}{n} = 0$ if $n > N$ or $n < 0$

Quick Examples

$$\binom{5}{2} = \frac{5!}{2!3!} = \frac{120}{2 \times 6} = 10 \quad \binom{10}{3} = \frac{10!}{3!7!} = \frac{10 \times 9 \times 8}{6} = 120 \quad \binom{52}{5} = 2,598,960$$

Classic Urn Sampling — Setup

Urn Model

An urn contains N balls: M **White** and $(N - M)$ **Red**.

Draw a sample of n balls *without replacement*, obtaining s white and $n - s$ red.

	White	Red	Total
Sample	s	$n - s$	n
Remaining	$M - s$	$(N - M) - (n - s)$	$N - n$
Total	M	$N - M$	N

Probability of Exactly s White Balls

$$P(S = s) = \frac{\binom{M}{s} \binom{N - M}{n - s}}{\binom{N}{n}}$$

This is the **Hypergeometric distribution**.

Urn Sampling — Clinical Trial Application

Comparing Two Therapies

N patients are randomised to Therapy A (M patients) or Therapy B ($N - M$ patients). Total successes observed: n .

	Success	Failure	Total
Therapy A	s	$M - s$	M
Therapy B	$n - s$	$(N - M) - (n - s)$	$N - M$
Total	n	$N - n$	N

Key Insight

If Therapy A is *not* active (null hypothesis), then the n total successes would be distributed among patients *by chance alone* — regardless of which therapy each patient received.

Under H_0 : all marginal totals are fixed, and

$$P(S = s) = \frac{\binom{M}{s} \binom{N-M}{n-s}}{\binom{N}{n}}$$

For example, $P(S = 2)$ gives the probability of observing exactly 2 successes in Therapy A by chance alone.

This is the basis of **Fisher's Exact Test**.

Combinations — Poker Hand Examples

5-card poker hands from a 52-card deck

Total hands: $\binom{52}{5} = 2,598,960$.

- ① **Four Aces:** Choose 4 aces $\binom{4}{4} = 1$, choose 1 from remaining 48: $\binom{48}{1} = 48$.
$$P = \frac{48}{2,598,960} \approx 0.0000185.$$
- ② **Four of a kind** (any rank): Choose rank (13), choose 4 of that rank $\binom{4}{4} = 1$, choose 1 remaining card (48): $13 \times 48 = 624$. $P \approx 0.000240$.
- ③ **Three of a kind** (exactly): Choose rank (13), choose 3 of 4 $\binom{4}{3} = 4$, choose 2 different ranks from remaining 12 $\binom{12}{2} = 66$, choose 1 suit each $4^2 = 16$: $13 \times 4 \times 66 \times 16 = 54,912$.
 $P \approx 0.0211$.

The Binomial Theorem

Binomial Theorem

For any real a, b and positive integer N :

$$(a + b)^N = \sum_{n=0}^N \binom{N}{n} a^n b^{N-n}$$

Special Cases

Case 1: $a = p$, $b = 1 - p$ (probabilities):

$$\sum_{n=0}^N \binom{N}{n} p^n (1-p)^{N-n} = 1$$

This shows that Binomial(N, p) probabilities sum to 1. ✓

Case 2: $a = b = 1$:

$$\sum_{n=0}^N \binom{N}{n} = 2^N$$

i.e. the total number of subsets of a set with N elements equals 2^N .

Combinations With Indistinguishable Objects

Two Types of Objects

Suppose we have two groups:

$A : \alpha_1, \alpha_2, \dots, \alpha_a$ (a indistinguishable objects)

$B : \beta_1, \beta_2, \dots, \beta_b$ (b indistinguishable objects)

How many distinct arrangements of all $a + b$ objects?

Formula

Consider $a + b$ spaces on a line. Choose a spaces for group A ; the remaining b spaces are filled by group B .

$$\text{Distinct arrangements} = \binom{a+b}{a} = \frac{(a+b)!}{a! b!}$$

Counting Summary Table

		Ordered	Unordered
Without replacement	(distinct objects)	$\frac{N!}{(N-r)!}$	$\binom{N}{r}$
With replacement	(repetition allowed)	N^r	$\binom{N+r-1}{r}$

Four Scenarios

Choose 3 from $\{A, B, C, D, E\}$ ($N = 5, r = 3$):

- Ordered, no rep: $P(5, 3) = 60$ (President, VP, Sec)
- Unordered, no rep: $\binom{5}{3} = 10$ (committee)
- Ordered, rep: $5^3 = 125$ (3-digit base-5 numbers)
- Unordered, rep: $\binom{7}{3} = 35$ (e.g. scoops of ice cream)

Probability via Counting – Poker Examples

5-Card Poker Hands

Total hands: $\binom{52}{5} = 2,598,960$.

Flush (5 same suit): $4 \times \binom{13}{5} = 5148$

$P(\text{flush}) = \frac{5148}{2598960} \approx 0.00198$

Full house (3+2): $13 \times \binom{4}{3} \times 12 \times \binom{4}{2} = 3744$

$P(\text{full house}) = \frac{3744}{2598960} \approx 0.00144$

4 of a kind: $13 \times \binom{4}{4} \times 48 = 624$

$P(4 \text{ of a kind}) = \frac{624}{2598960} \approx 0.00024$

From Counting to Formal Probability

Classical approach: works only when outcomes are equally likely.

Need a more general framework.

Historical Note

Andrey Kolmogorov (1933) published *Grundbegriffe der Wahrscheinlichkeitsrechnung* (“Foundations of Probability Theory”), providing the axiomatic basis still used today.

Probability Function

A **probability function** P assigns a real number to each event in a σ -algebra $\mathcal{B}$ of subsets of S . It must satisfy three axioms.

σ -Algebra (Borel Field)

σ -Algebra

A collection $\mathcal{B}$ of subsets of S is a σ -**algebra** if:

- ① $\emptyset \in \mathcal{B}$
- ② $A \in \mathcal{B} \Rightarrow A^c \in \mathcal{B}$ (closed under complement)
- ③ $A_1, A_2, \dots \in \mathcal{B} \Rightarrow \bigcup_{i=1}^{\infty} A_i \in \mathcal{B}$ (closed under countable union)

Examples:

- Trivial: $\mathcal{B} = \{\emptyset, S\}$
- Full: $\mathcal{B} = 2^S$ (all subsets)
- Borel σ -algebra on $\mathbb{R}$: generated by all open intervals (a, b)

Why σ -algebras?

For $S = \mathbb{R}$, not all subsets can be assigned a probability consistently (non-measurable sets exist). The σ -algebra defines the *admissible events*.

Kolmogorov's Three Axioms

Axioms of Probability (Kolmogorov 1933)

Given $(S, \mathcal{B})$, a function $P : \mathcal{B} \rightarrow \mathbb{R}$ is a **probability measure** if:

(A1) $P(E) \geq 0$ for all $E \in \mathcal{B}$ (*non-negativity*)

(A2) $P(S) = 1$ (*normalisation*)

(A3) If $E_1, E_2, \dots$ are **pairwise disjoint** events, then

$$P\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} P(E_i) \quad (\text{countable additivity})$$

The triple $(S, \mathcal{B}, P)$ is called a **probability space**.

Constructing Probabilities

Theorem (CB 1.2.6)

Let $S = \{s_1, \dots, s_n\}$ be finite (or countably infinite). Let $p_1, \dots, p_n \geq 0$ with $\sum_i p_i = 1$. Define $P(A) = \sum_{\{i: s_i \in A\}} p_i$. Then P is a probability.

Coin Toss

$S = \{H, T\}$, $p_H = p_T = 1/2$.
 $P(\{H\}) = 1/2$, $P(S) = 1$. ✓

Biased Die

$S = \{1, \dots, 6\}$,
 $P(\{i\}) = p_i$, $\sum p_i = 1$.
Fair: $p_i = 1/6$ for all i .

Urn Model

w white + r red balls, draw 1.

$$P(\text{white}) = \frac{w}{w+r},$$

$$P(\text{red}) = \frac{r}{w+r}.$$

Follows from equally likely outcomes and fundamental theorem.

Hypergeometric: Urn Model and Fisher's Exact Test

Classic Urn Sampling

Urn: N balls, M white, $N - M$ red. Draw n without replacement. $Y = \# \text{white}$.

Sample space: $\max(0, n - (N - M)) \leq Y \leq \min(n, M)$

$$P(Y = s) = \frac{\binom{M}{s} \binom{N-M}{n-s}}{\binom{N}{n}}$$

Vaccine Quality Control

Batches of N vials; M defective. Sample n vials. Accept if 0 defective.

$$P(\text{accept}) = \frac{\binom{M}{0} \binom{N-M}{n}}{\binom{N}{n}} \approx \left(1 - \frac{M}{N}\right)^n \quad (\text{for large } N)$$

Table of $P(\text{accept})$: defect rate 0.1%, 1%, 10%; sample size 5, 10, 50.

In-Class Activity – Lecture 3

Activity 3 – Combinations & Axioms

Part A:

- Ⓐ A class has 8 women and 5 men. A committee of 4 is chosen at random. Find $P(\text{exactly 2 women})$.
- Ⓑ A repair facility has 25 keyboards: 6 electrical (E), 19 mechanical (M) defects. 5 are chosen at random.
 - (i) $P(\text{exactly 2 electrical defects})$
 - (ii) $P(\text{at most 1 electrical defect})$

Part B – Axioms:

- Ⓒ Which of the following can be a valid σ -algebra on $S = \{1, 2, 3\}$? (i) $\{\emptyset, \{1\}, \{2, 3\}, S\}$ (ii) $\{\emptyset, \{1\}, S\}$ (iii) All subsets of S
- Ⓓ For $S = \{1, 2, 3, 4\}$, define $P(\{1\}) = 0.1$, $P(\{2\}) = 0.2$, $P(\{3\}) = 0.3$, $P(\{4\}) = 0.4$. Verify that this defines a valid probability and find $P(\{1, 3\})$.

Activity 3 – Solutions

Part A:

Ⓐ Total: $\binom{13}{4} = 715$. Favorable: $\binom{8}{2} \binom{5}{2} = 28 \times 10 = 280$.
 $P = 280/715 \approx \mathbf{0.392}$

Ⓑ (i) $\frac{\binom{6}{2} \binom{19}{3}}{\binom{25}{5}} = \frac{15 \times 969}{53130} = \frac{14535}{53130} \approx \mathbf{0.274}$

(ii) $\frac{\binom{6}{0} \binom{19}{5} + \binom{6}{1} \binom{19}{4}}{\binom{25}{5}} = \frac{11628 + 29070}{53130} = \frac{40698}{53130} \approx \mathbf{0.766}$

Part B:

Ⓒ (i) **Valid:** $\emptyset \checkmark$, closed under c : $\{1\}^c = \{2, 3\} \checkmark$, $S^c = \emptyset \checkmark$; closed under $\cup \checkmark$.

(ii) **Not valid:** $\{1\}^c = \{2, 3\} \notin \mathcal{B}$.

(iii) **Valid:** power set always a σ -algebra.

Ⓓ $\sum p_i = 0.1 + 0.2 + 0.3 + 0.4 = 1 \checkmark$; all $p_i \geq 0 \checkmark$. $P(\{1, 3\}) = 0.1 + 0.3 = \mathbf{0.4}$.